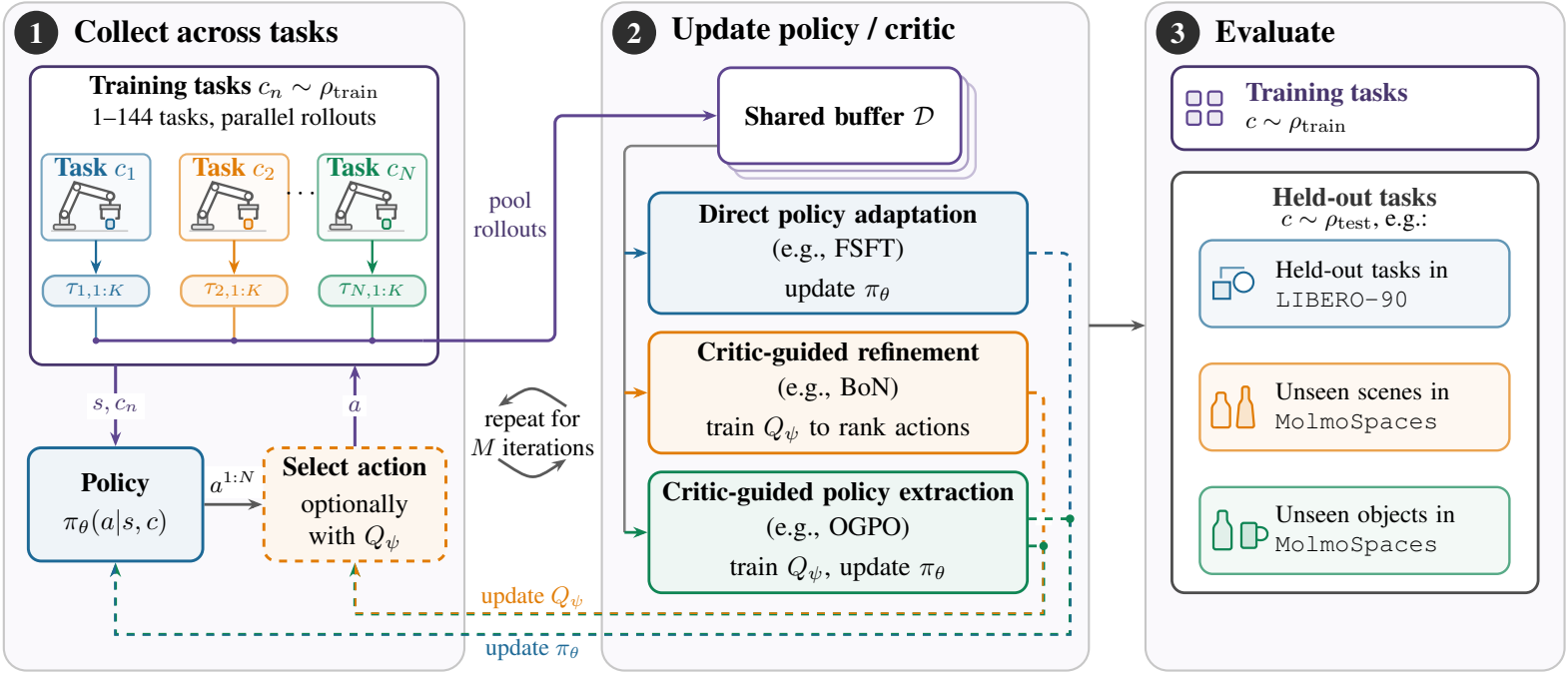

1 Collect across tasks
Training tasks $c_n \sim \rho_{\text{train}}$
1–144 tasks, parallel rollouts
Task c_1
Task c_2
Task c_N
$\tau_{1,1:K}$
$\tau_{2,1:K}$
$\tau_{N,1:K}$
pool rollouts
s, c_n
a
Policy
$\pi_\theta(a|s,c)$
$a^{1:N}$
Select action
optionally with Q_ψ
repeat for M iterations
2 Update policy / critic
Shared buffer $\mathcal{D}$
Direct policy adaptation
(e.g., FSFT)
update π_θ
Critic-guided refinement
(e.g., BoN)
train Q_ψ to rank actions
Critic-guided policy extraction
(e.g., OGPO)
train Q_ψ, update π_θ
update Q_ψ
update π_θ
3 Evaluate
Training tasks
$c \sim \rho_{\text{train}}$
Held-out tasks
$c \sim \rho_{\text{test}}$, e.g.:
Held-out tasks in LIBERO-90
Unseen scenes in MolmoSpaces
Unseen objects in MolmoSpaces